FOUR EQUIVALENT FORMULATIONS OF THE J-2 PROPERTY FOR A LOCALLY NOETHERIAN SCHEME, AND THE TRANSFER TO SCHEMES LOCALLY OF FINITE TYPE OVER IT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-J is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 3309-3323 and proof span 3325-3354 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. Reread under the 2026-10-14 prefilter rule (prefilter hit is not a rejection when proof_bytes is at least 800 or the proof runs at least 100 lines): this candidate is 1341 bytes over 30 proof lines and the prefilter matched the string 'by definition' at proof line 3337. On rereading, that occurrence is a citation of the definition of a J-2 ring (More on Algebra, Definition more-algebra-definition-J), not an author-declared omission, and the surrounding text is an argument; the candidate was therefore accepted rather than skipped.

Editorial note. Tag self-check per LANE_BRIEF 10.5 (5): the label lemma-J resolves in the project's own tag list to canonical tag 07R4 (line 9854 of the index, full label morphisms-lemma-J). The same-chapter public tag id set is empty because reference/public-items.jsonl holds no row whose primary source_file is morphisms.tex, so the intersection of the resolved tag id with the public tag ids of this chapter is the empty set and the candidate passes. The gate was run as: python operations/agent-authoring/tools/precard_gate.py –source morphisms.tex –label lemma-J –start 3309 –end 3323 –proof-start 3325 –proof-end 3354 –local-id 34087, giving RESULT=OK with all four span guards passing, no public interval row for this chapter, and occupancy FREE for morphisms.tex 3309 3354.

Editorial note. Spans were recomputed live by label location in the pinned chapter file (locate.py) before the card was written, so the statement span 3309-3323 and the proof span 3325-3354 are the real begin/end environment lines on the current disk; the byte counts of the located text were checked against the pool candidate (statement 432 characters, proof 1135 characters after whitespace removal, matching the guard report).

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Editorial note. 4 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- An open immersion is a morphism locally of finite type, cited from the same chapter. Cited at proof line 3326. Lets an open subscheme of a J-2 scheme be treated as a scheme locally of finite type over it, which is what the transfer assertion of the lemma needs.
- A composition of morphisms locally of finite type is locally of finite type, cited from the same chapter. Cited at proof lines 3328-3329. Allows a scheme locally of finite type over an open of the base to be viewed as locally of finite type over the base itself.
- The definition of a J-2 ring, in the form that every finite type algebra over it has open regular locus, cited from the more algebra chapter. Cited at proof lines 3335-3338. Supplies the ring-level content of the J-2 condition, which is used in both directions of the equivalence.
- A morphism locally of finite type is characterized by finite type ring maps on affine opens, cited from the same chapter. Cited at proof lines 3347-3350. Turns local finiteness of type of the morphism into finiteness of type of the induced ring maps, so that the regular locus statement becomes available on affine pieces.

Source reference. Original lemma 01T3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2360–2360. Same-chapter reference used in the retained author text at source line(s) 2889, 2937, 3138, 3329, 9763, 9913. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01T5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2362–2362. Same-chapter reference used in the retained author text at source line(s) 2980, 3135, 3326, 9944. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01T2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2359–2359. Same-chapter reference used in the retained author text at source line(s) 2717, 2739, 2801, 2863, 3019, 3035, 3157, 3348, 3613, 3707, 3868, 5197, 5282, 6702, 9325, 9390, 9440, 11421, 13422, 13686. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original definition 07R3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9853–9853. Same-chapter reference used in the retained author text at source line(s) 3307, 3338. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let X be a locally Noetherian scheme. The following are equivalent*

(1) *X is J-2,*
(2) *there exists an open covering of X all of whose members are J-2 schemes,*
(3) *for every affine open $\mathrm{Spec}(R) = U \subset X$ the ring R is J-2, and*

(4) *there exists an affine open covering $S = \bigcup U_i$ such that each $\mathcal{O}(U_i)$ is J-2 for all i.*

Moreover, in this case any scheme locally of finite type over X is J-2 as well.

Proof. By Lemma 01T5 an open immersion is locally of finite type. A composition of morphisms locally of finite type is locally of finite type (Lemma 01T3). Thus it is clear that if X is J-2 then any open and any scheme locally of finite type over X is J-2 as well. This proves the final statement of the lemma.

If $\mathrm{Spec}(R)$ is J-2, then for every finite type R-algebra A the regular locus of the scheme $\mathrm{Spec}(A)$ is open. Hence R is J-2, by definition (see More on Algebra, Definition 07R3). Combined with the remarks above we conclude that (1) implies (3), and (2) implies (4). Of course (1) $\Rightarrow$ (2) and (3) $\Rightarrow$ (4) trivially.

To finish the proof we show that (4) implies (1). Assume (4) and let $Y \to X$ be a morphism locally of finite type. We can find an affine open covering $Y = \bigcup V_j$ such that each $V_j \to X$ maps into one of the U_i. By Lemma 01T2 the induced ring map $\mathcal{O}(U_i) \to \mathcal{O}(V_j)$ is of finite type. Hence the regular locus of $V_j = \mathrm{Spec}(\mathcal{O}(V_j))$ is open. Since $\mathrm{Reg}(Y) \cap V_j = \mathrm{Reg}(V_j)$ we conclude that $\mathrm{Reg}(Y)$ is open as desired. $\square$